

练习题-定积分计算

1. 计算下列定积分

$$(1) \int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx; \quad (2) \int_0^1 \frac{dx}{(1+x)\sqrt{1+x^2}}$$

2. 已知 $f(0)=1, f(2)=3, f'(2)=5$, 计算 $\int_0^1 xf''(2x)dx$.

3. 已知 $y' = \arctan(x-1)^2, y(0)=0$, 求 $\int_0^1 y(x)dx$.

4. 计算积分 $\int_0^1 xf(x)dx$, 其中 $f(x) = \int_1^{x^2} \frac{\sin x}{x} dx$.

5. 已知 $f(x) = \int_0^x e^{(x-t)^2} dt$, 求 $\int_0^1 [f(x) - f(1)] dx$.

6. 计算下列定积分

$$(1) \int_{-2}^2 [e^{-x}(1+e^x)^2 \sin^3 x + x^2] dx; \quad (2) \int_{-2}^2 [\sin^3 x + x \ln(1+e^x)] dx;$$

$$(3) \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\sin^4 x + x^2 \ln \frac{2-x}{2+x} \right] dx; \quad (4) \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan e^{-x} dx;$$

$$(5) \int_{\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\ln|x + \sqrt{1+x^2}| + \frac{e^x}{1+e^x} \right] \sin^2 x dx.$$

7. 计算下列定积分

$$(1) \int_0^{\frac{\pi}{2}} \frac{\ln(1+\sqrt{\sin x}) - \ln(1+\sqrt{\cos x}) + \sin^3 x}{2} dx; \quad (2) \int_0^2 \frac{4-x^2}{x+\sqrt{4-x^2}} dx;$$

$$(2) \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx; \quad (4) \int_0^{\pi} \frac{1}{1+a \cos x} dx \quad (0 < a < 1);$$

$$(5) \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx.$$

$$8. \text{ 计算 } \int_{e^{-2n\pi}}^1 \left[\cos\left(\ln \frac{1}{x}\right) \right]' dx.$$

9. 设 $f(x) = \int_1^x \frac{\ln x}{1+x} dx$, 求 $f(x) + f\left(\frac{1}{x}\right)$.

10. 设 $f(x) = \begin{cases} xe^{-x^2}, & x \geq 0 \\ \frac{1}{1 + \cos x}, & -1 \leq x < 0 \end{cases}$, 计算 $\int_1^4 f(x-2)dx$.

11. 设 $f(x) = \begin{cases} 2x + \frac{3}{2}x^2, & -1 \leq x < 0 \\ \frac{xe^x}{(1+e^x)^2}, & 0 \leq x \leq 1 \end{cases}$, 求 $F(x) = \int_{-1}^x f(t)dt$ 的表达式.

12. 设 $f(x) = x, x \geq 0, g(x) = \begin{cases} \sin x, & 0 \leq x \leq \frac{\pi}{2} \\ 0, & x > \frac{\pi}{2} \end{cases}$, 分别求 $0 \leq x \leq \frac{\pi}{2}$ 与 $x > \frac{\pi}{2}$ 时积

分 $\int_0^x f(t)g(x-t)dt$ 的表达式.

13. 计算积分 $\int_0^{n\pi} x|\sin x|dx$ ($n=1,2,\dots$).

14. 已知 $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases}$, 计算 $S_n = \int_{2n}^{2n+2} f(x-2n)e^{-x}dx$ ($n=0,1,\dots$),

并计算极限 $\lim_{n \rightarrow \infty} (S_0 + S_1 + \dots + S_n)$.